

By applying the "**What? / So What? / Now What?**" framework to a specifications grading model, we shift the focus from "earning points" to demonstrating a logical progression of thought.

In this system, a student must complete **all** the requirements in a bundle to earn that grade.

Critical Reflection: "Three-Stage Model" Rubric

Bundle C: The Foundational Level

Demonstrates basic engagement with the model.

- **The "What":** Clearly identifies a specific problem or issue encountered during the first half of the summer research project. Provides a chronological account of what happened and what was observed.
 - **The "So What":** States at least one reason why the experience matters to the student or the project.
 - **The "Now What":** Identifies one lesson learned and provides a general statement about how they will move forward.
 - **Format:** Meets the minimum word count (400-500 words) and is free of major grammatical errors that hinder understanding.
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Bundle B: The Practitioner Level

Includes all requirements from Bundle C, plus the following:

- **Analysis (So What):** Moves beyond personal feeling to identify **critical questions** raised by the experience. Draws at least one logical conclusion based on the evidence of what happened.
 - **Application (Now What):** Describes a **specific action** or change in behavior that will be implemented as a result of this insight.
 - **Context:** Explicitly links the research problem to the broader goals of the summer project or the field of study.
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Bundle A: The Researcher Level

Includes all requirements from Bundles C and B, plus the following:

- **Synthesis:** Successfully connects the "What" to theoretical concepts or research methodologies discussed in lab, or coursework, or literature.
 - **Depth of Perception:** The "So What" section demonstrates significant personal growth or a shift in perspective regarding how research is conducted.
 - **Transformative Learning:** The "Now What" section explains how their **fundamental approach** to learning or problem-solving has changed based on this experience (i.e., "I used to think X, but now I understand Y, so I will approach Z differently").
 - **Professionalism:** Writing is polished, showing a high degree of self-awareness and critical distance.
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The "Unsatisfactory / Revise" Category

The reflection is marked as **Unsatisfactory** if it meets any of the following:

- **Surface-Level Only:** Fails to move past the "What" stage (purely descriptive with no analysis).
- **Incomplete:** One of the three stages (What / So What / Now What) is missing or addressed in only 1–2 sentences.
- **Vague:** Claims "learning occurred" without providing specific examples or evidence from the research project.
- **Disorganized:** The reflection does not follow the three-stage structure, making the progression of thought difficult to track.

Grading Key

Bundle Completed	Final Grade
Bundle A	90+
Bundle B	80-90
Bundle C	70-80
Does not meet Bundle C	<70

